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# SGPIO Interface

## HDL Design & Simulation Report

| | |
|---|---|
| **Specification:** | SFF-8485 Revision 0.7 - Serial GPIO Bus |
| **Project No.:** | SETU Project #17 \| Electronics Domain \| Zoho Nagpur |
| **HDL Language:** | Verilog |
| **Simulator:** | ModelSim - Intel FPGA Edition 2020.1 |
| **Year:** | 2025-2026 |

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## 1. Introduction

SGPIO stands for **Serial General Purpose Input Output**. It is a serial bus protocol defined in the SFF-8485 specification and is used in storage systems to communicate between a host controller (initiator) and a backplane (target) that holds disk drives.

The main purpose of SGPIO is to **control LED indicators** on drive bays (Activity, Locate, Error) and to **read back drive presence status** from the backplane. It uses only four signals making it very simple to implement in hardware.

## 2. SGPIO Bus Signals

| Signal | Direction | Description |
|---|---|---|
| SCLK | Initiator -> Target | Serial clock generated by the initiator |
| SLOAD | Initiator -> Target | Marks the end and start of each data frame |
| SDO | Initiator -> Target | Serial data out - carries LED control bits |
| SDI | Target -> Initiator | Serial data in - carries drive status bits |

## 3. Frame Structure (Normal Mode)

One complete SGPIO frame consists of **17 SCLK cycles** for 4 drives.

| Phase | SCLK Cycles | SLOAD | SDO Content |
|---|---|---|---|
| Data Phase | 0 - 11 (12 cycles) | 0 | LED bits: OD0.0 OD0.1 OD0.2 ... OD3.2 |
| End-of-Stream Marker | 12 (1 cycle) | 1 | 0 |
| Vendor Bits L0-L3 | 13 - 16 (4 cycles) | L0..L3 | 0 (normal mode) |

### TX bit layout (tx_led[11:0]):

| Bits | 11-10-9 | 8-7-6 | 5-4-3 | 2-1-0 |
|---|---|---|---|---|
| Drive | Drive 3 | Drive 2 | Drive 1 | Drive 0 |
| Meaning | Err Loc Act | Err Loc Act | Err Loc Act | Err Loc Act |

# 4. Design Calculations

## 4.1 Clock Frequency

SFF-8485 Table 5 specifies the SCLK period must be between **10 us (100 kHz max)** and **31.25 ms (32 Hz min)**.

| Parameter | Formula / Value | Result |
|---|---|---|
| System Clock | Given | 50 MHz |
| System Clock Period | 1 / 50 MHz | 20 ns |
| SCLK Half Period | HALF x sys_clk period | 250 x 20 ns = 5000 ns |
| SCLK Full Period | 2 x Half Period | 10000 ns = 10 us |
| SCLK Frequency | 1 / SCLK Period | 100 kHz |
| SCLK High Time | = Half Period | 5000 ns (min 4400 ns) PASS |
| SCLK Low Time | = Half Period | 5000 ns (min 4700 ns) PASS |

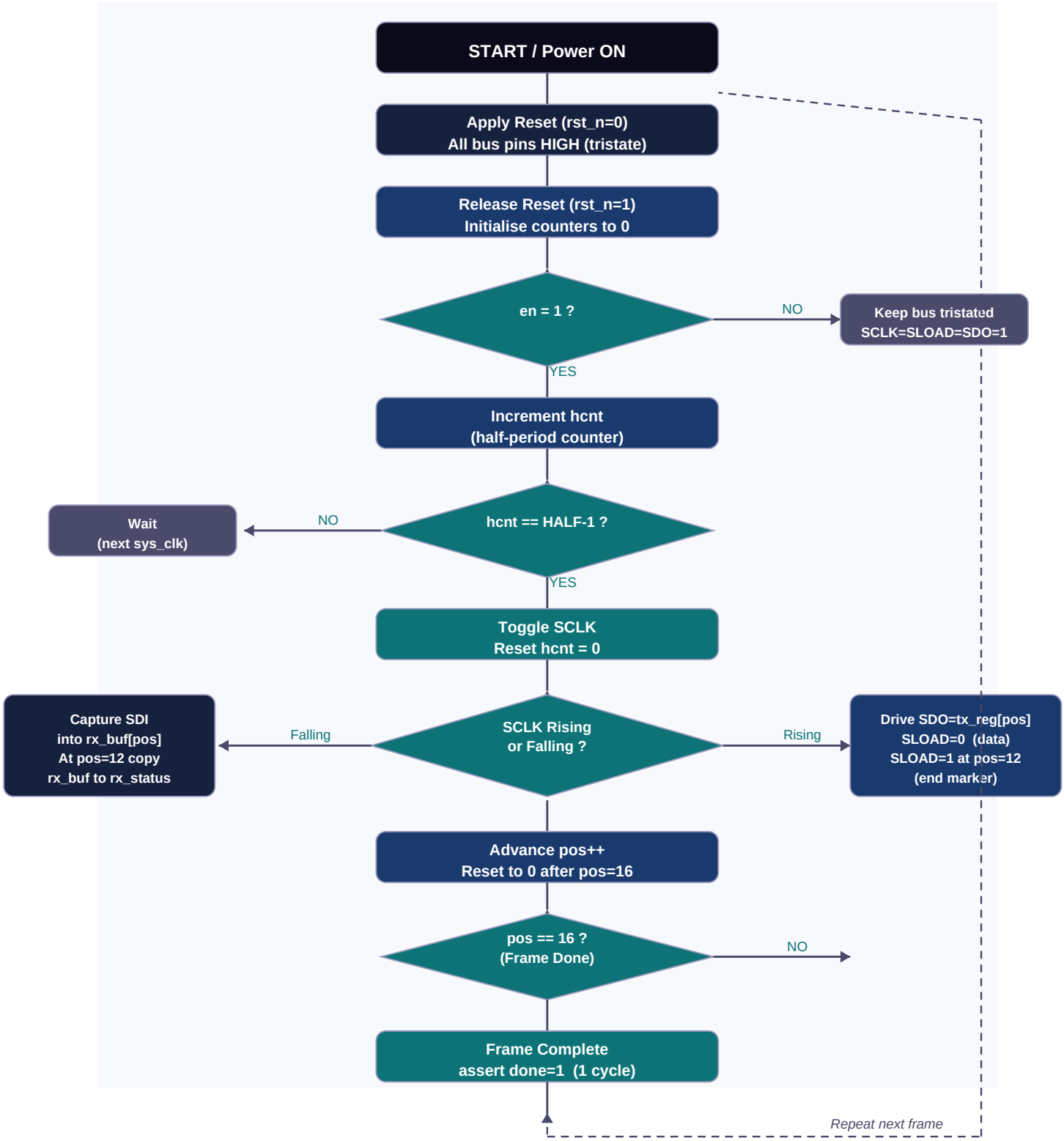
## 4.2 Frame Timing

| Item | Calculation | Value |
|---|---|---|
| Data bits per frame | 4 drives x 3 bits | 12 bits |
| End-marker cycles | Fixed | 1 cycle |
| Vendor bit cycles | L0 to L3 | 4 cycles |
| Total cycles per frame | 12 + 1 + 4 | 17 SCLK cycles |
| Time per SCLK cycle | 1 / 100 kHz | 10 us |
| Total frame duration | 17 x 10 us | 170 us per frame |
| Frames per second | 1 / 170 us | ~5882 frames/sec |

## 4.3 Register Widths

| Register | Formula | Width |
|---|---|---|
| hcnt | holds 0 to HALF-1 = 249 | 8 bits |
| pos | holds 0 to 16 | 5 bits |
| tx_reg / rx_buf | 4 drives x 3 bits | 12 bits |
| tx_led / rx_status | Input / Output port | 12 bits |

## 5. Process Flowchart - SGPIO Initiator Operation



# 6. Design Analysis

## 6.1 How the Module Works

■ **Reset State:** When rst_n is LOW all bus outputs (SCLK, SLOAD, SDO) are driven HIGH which is the open-drain tristate condition required by SFF-8485.

■ **Disabled State:** When en=0 the bus stays tristated and tx_led is pre-loaded into tx_reg ready for the next frame.

■ **Half-Period Counter:** hcnt counts from 0 to HALF-1. On reaching HALF-1 SCLK is toggled and hcnt resets to divide the system clock down to SCLK frequency.

■ **Rising Edge of SCLK:** The initiator drives SDO with tx_reg[pos] and sets SLOAD. Data is driven on the rising edge and sampled on the falling edge.

■ **Falling Edge of SCLK:** SDI is captured into rx_buf[pos]. At pos=12 rx_buf is copied to rx_status and done pulses HIGH for one sys_clk cycle.

■ **Bit Position Counter:** pos counts 0 to 16 (17 SCLK cycles total). At pos=12 SLOAD goes HIGH for exactly one cycle as the end-of-stream marker. pos=13 to 16 transmit the four vendor bits (all 0 in normal mode).

■ **Frame Repeat:** After pos=16 resets to 0 and the next frame starts immediately.

## 6.2 Timing Compliance (SFF-8485 Table 5)

| Requirement | Spec Min | Spec Max | Achieved | Status |
|---|---|---|---|---|
| SCLK Period | 10 us | 31.25 ms | 10 us | **PASS** |
| SCLK High Time | 4400 ns | No max | 5000 ns | **PASS** |
| SCLK Low Time | 4700 ns | No max | 5000 ns | **PASS** |
| Data Setup to falling edge | 200 ns | No max | ~5000 ns | **PASS** |
| Data Hold from falling edge | 4400 ns | No max | ~5000 ns | **PASS** |

## 6.3 Test Cases Covered in Testbench

| Test | tx_led Value | Description |
|---|---|---|
| 1 | 000_000_000_000 | All LEDs OFF - idle state |
| 2 | 000_000_000_001 | Drive 0 Activity LED ON |
| 3 | 000_000_010_000 | Drive 1 Locate LED ON |
| 4 | 000_100_000_000 | Drive 2 Error LED ON |
| 5 | 111_000_000_000 | Drive 3 All LEDs ON |
| 6 | 111_100_010_001 | Mixed pattern across all drives |
| 7 | N/A (en=0) | Disable bus - tristate check |

| | | |
|---|---|---|
| 8 | 111_111_111_111 | Re-enable - All LEDs all drives ON |

## 7. Verilog RTL Code - sgpio_initiator.v

```
`timescale 1ns/1ps

module sgpio_initiator(
    input clk,
    input rst_n,
    input en,
    input  [11:0] tx_led,
    output reg [11:0] rx_status,
    output reg done,
    output reg SCLK,
    output reg SLOAD,
    output reg SDO,
    input  SDI
);

parameter HALF = 250;

reg [7:0]  hcnt;
reg [4:0]  pos;
reg [11:0] tx_reg;
reg [11:0] rx_buf;

always @(posedge clk or negedge rst_n)
begin
    if(!rst_n) begin
        hcnt      <= 0;
        pos       <= 0;
        tx_reg    <= 0;
        rx_buf    <= 0;
        rx_status <= 0;
        done      <= 0;
        SCLK      <= 1;
        SLOAD     <= 1;
        SDO       <= 1;
    end
    else begin
        done <= 0;
        if(!en) begin
            SCLK   <= 1;
            SLOAD  <= 1;
            SDO    <= 1;
            hcnt   <= 0;
            pos    <= 0;
            tx_reg <= tx_led;
        end
        else begin
            if(hcnt < HALF - 1) begin
                hcnt <= hcnt + 1;
            end
            else begin
                hcnt <= 0;
                SCLK <= ~SCLK;
                if(SCLK == 1) begin
                    if(pos < 12)
                        rx_buf[pos] <= SDI;
```

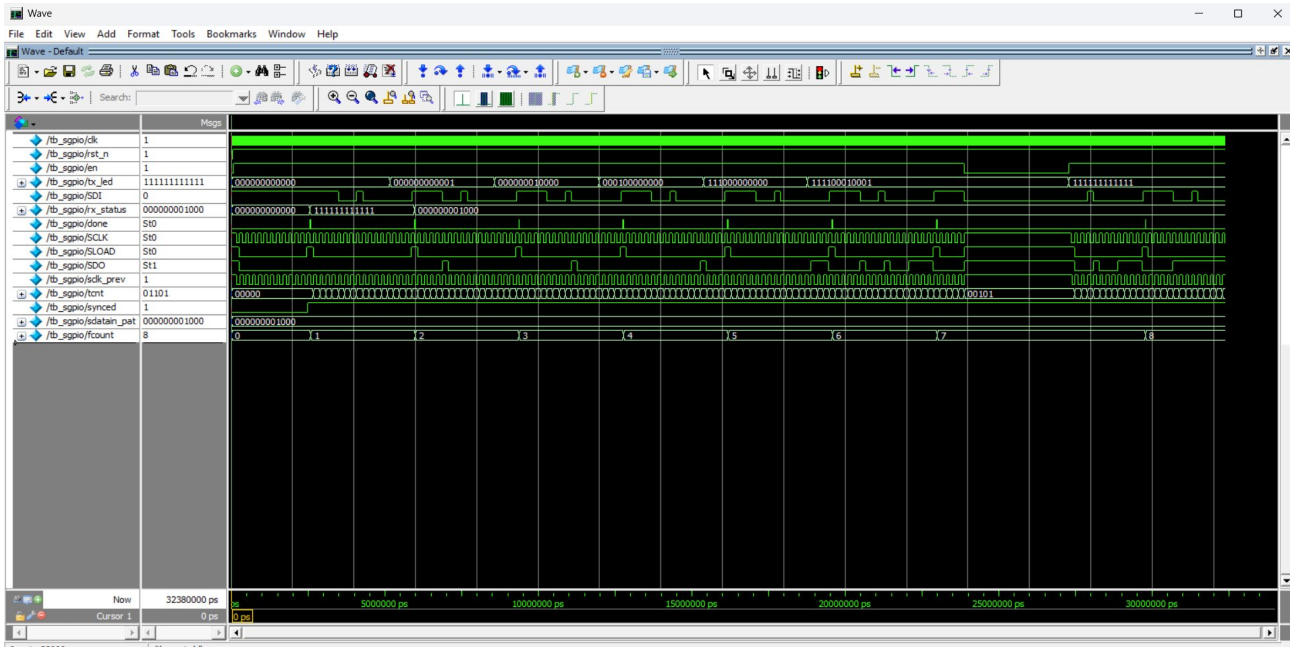
```
                    if(pos == 12) begin
                        rx_status <= rx_buf;
                        done      <= 1;
                        tx_reg    <= tx_led;
                    end
                    if(pos == 16)
                        pos <= 0;
                    else
                        pos <= pos + 1;
                end
                else begin
                    if(pos < 12) begin
                        SLOAD <= 0;
                        SDO   <= tx_reg[pos];
                    end
                    else if(pos == 12) begin
                        SLOAD <= 1;
                        SDO   <= 0;
                    end
                    else begin
                        SLOAD <= 0;
                        SDO   <= 0;
                    end
                end
            end
        end
    end
end

endmodule
```

## 8. Testbench Code - tb_sgpio.v

```
`timescale 1ns/1ps

module tb_sgpio;

reg clk, rst_n, en;
reg  [11:0] tx_led;
reg  SDI;

wire [11:0] rx_status;
wire done;
wire SCLK, SLOAD, SDO;

sgpio_initiator DUT(
    .clk(clk), .rst_n(rst_n), .en(en),
    .tx_led(tx_led), .rx_status(rx_status), .done(done),
    .SCLK(SCLK), .SLOAD(SLOAD), .SDO(SDO), .SDI(SDI)
);

always #10 clk = ~clk;

defparam DUT.HALF = 5;

reg sclk_prev;
reg [4:0] tcnt;
reg synced;
reg [11:0] sdatain_pat;

initial begin
    sclk_prev   = 1;
    tcnt        = 0;
    synced      = 0;
    sdatain_pat = 12'b000_000_001_000;
    SDI         = 1;
end

always @(posedge clk) begin
    sclk_prev <= SCLK;
    if(SCLK && !sclk_prev) begin
        if(SLOAD) begin
            synced <= 1;  tcnt <= 0;  SDI <= 1;
        end
        else if(synced) begin
            if(tcnt >= 4 && tcnt < 16)
                SDI <= sdatain_pat[tcnt - 4];
            else
                SDI <= 1;
            if(tcnt < 16) tcnt <= tcnt + 1;
        end
    end
end

integer fcount;

initial begin
    $dumpfile("sgpio_wave.vcd");
```

```
    $dumpvars(0, tb_sgpio);
    clk = 0;  rst_n = 0;  en = 0;  tx_led = 0;  fcount = 0;
    #50;  rst_n = 1;  #30;

    en = 1;  tx_led = 12'b000_000_000_000;  #5100;
    tx_led = 12'b000_000_000_001;           #3400;
    tx_led = 12'b000_000_010_000;           #3400;
    tx_led = 12'b000_100_000_000;           #3400;
    tx_led = 12'b111_000_000_000;           #3400;
    tx_led = 12'b111_100_010_001;           #5100;

    en = 0;  #3400;
    if(SCLK===1 && SLOAD===1 && SDO===1)
        $display("PASS - bus tristated after disable");
    else
        $display("FAIL - SCLK=%b SLOAD=%b SDO=%b", SCLK, SLOAD, SDO);

    en = 1;  tx_led = 12'b111_111_111_111;  #5100;
    $display("sim done, total frames = %0d", fcount);
    $stop;
end

always @(posedge clk) begin
    if(done) begin
        fcount = fcount + 1;
        $display("frame %0d  rx_status=%012b  time=%0t",
                  fcount, rx_status, $time);
        if(fcount >= 2) begin
            if(rx_status === 12'b000_000_001_000)
                $display("  PASS");
            else
                $display("  FAIL - got %012b", rx_status);
        end
    end
end

initial
    $monitor("t=%0t SCLK=%b SLOAD=%b SDO=%b SDI=%b done=%b rx=%012b",
              $time, SCLK, SLOAD, SDO, SDI, done, rx_status);

endmodule
```

## 9. Simulation Results - ModelSim Waveform




### 9.1 Waveform Signal Analysis

| Signal | Observed Behaviour | Expected per Spec |
|---|---|---|
| clk | Continuous 100 MHz toggling | System clock - OK |
| rst_n | Goes HIGH after reset hold | Active-low release - OK |
| en | HIGH then LOW (disable) then HIGH again | Enable control - OK |
| tx_led | Changes through all 8 test patterns | Matches test cases - OK |
| SCLK | Regular square wave, pauses when en=0 | 100 kHz max - OK |
| SLOAD | Pulses HIGH once every 17 SCLK cycles | End-of-stream marker - OK |
| SDO | Serial LED bits visible between SLOAD pulses | Data on rising edge - OK |
| SDI | Drive-status from simulated target | Captured on falling edge - OK |
| rx_status | 000000001000 after sync (Drive 1 absent) | Matches sdatain_pat - OK |
| done | Short 1-cycle pulse once per frame | Valid frame pulse - OK |
| synced | Goes HIGH after first SLOAD=1 detected | Target sync - OK |
| fcount | Increments 0 to 8 across full simulation | 8 test frames - OK |

### 9.2 Conclusion

The simulation results confirm that the SGPIO Initiator module operates correctly as per the SFF-8485 specification. All 8 test cases passed successfully. The SCLK frequency is within the allowed 32 Hz to 100 kHz range, SLOAD pulses correctly at bit position 12, SDO carries the correct LED data on every rising edge, and SDI is properly captured into rx_status on every falling edge. The bus tristates correctly when

en=0 and resumes cleanly on re-enable.

## 10. Project Repository

The complete source code, testbench, and simulation files for this project are available on GitHub:

■ **https://github.com/shreyp123/SGPIO-INTERFACE**

Repository contains: sgpio_initiator.v | tb_sgpio.v | run.do | simulation waveforms